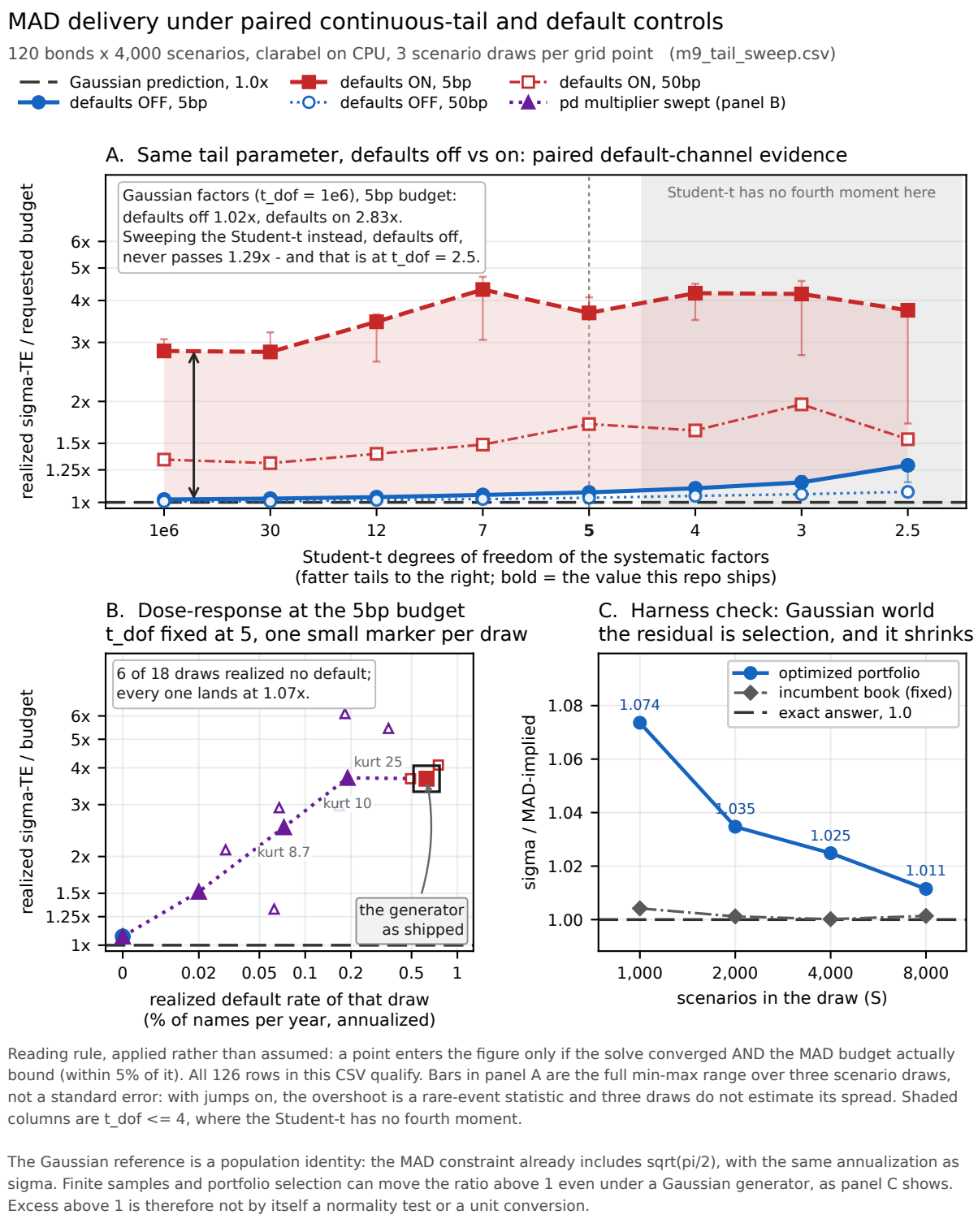
MAD delivery under paired continuous-tail and default controls

120 bonds x 4,000 scenarios, clarabel on CPU, 3 scenario draws per grid point (m9_tail_sweep.csv)



Reading rule, applied rather than assumed: a point enters the figure only if the solve converged AND the MAD budget actually bound (within 5% of it). All 126 rows in this CSV qualify. Bars in panel A are the full min-max range over three scenario draws, not a standard error: with jumps on, the overshoot is a rare-event statistic and three draws do not estimate its spread. Shaded columns are t_dof <= 4, where the Student-t has no fourth moment.

The Gaussian reference is a population identity: the MAD constraint already includes sqrt(pi/2), with the same annualization as sigma. Finite samples and portfolio selection can move the ratio above 1 even under a Gaussian generator, as panel C shows. Excess above 1 is therefore not by itself a normality test or a unit conversion.